

MA2031- Linear Algebra for Engineers

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Revision (Modules 1-4)

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Linear transformation and matrix representation

- Suppose $\mathcal{V}, \mathcal{W}$ are $\mathbb{F}$ -vector spaces. Then $T : \mathcal{V} \rightarrow \mathcal{W}$ is said to be a **linear transformation** if

$$T(\alpha v_1 + v_2) = \alpha T(v_1) + T(v_2) \quad \forall v_1, v_2 \in \mathcal{V}, \alpha \in \mathbb{F}.$$

- Fix basis $\mathcal{B}, \mathcal{E}$ for $\mathcal{V}, \mathcal{W}$ respectively. Assume $\dim(\mathcal{V}) = m, \dim(\mathcal{W}) = n$. Then $\varepsilon[\cdot]_{\mathcal{B}} : L(\mathcal{V}, \mathcal{W}) \rightarrow M_{n \times m}(\mathbb{F})$ is an isomorphism.

$$T \mapsto \varepsilon[T]_{\mathcal{B}}$$

(Recall: $[T(v)]_{\mathcal{E}} = \varepsilon[T]_{\mathcal{B}} [v]_{\mathcal{B}}$ for all $v \in \mathcal{V}$).

- Suppose $\mathcal{V} = \mathcal{W}$ and $\mathcal{B} = \mathcal{E}$. Then under the above isomorphism

$$S + T \mapsto {}_{\mathcal{B}}[S + T]_{\mathcal{B}} = {}_{\mathcal{B}}[S]_{\mathcal{B}} + {}_{\mathcal{B}}[T]_{\mathcal{B}}$$

$$\alpha T \mapsto {}_{\mathcal{B}}[\alpha T]_{\mathcal{B}} = \alpha {}_{\mathcal{B}}[T]_{\mathcal{B}}$$

$$S \circ T \mapsto {}_{\mathcal{B}}[S \circ T]_{\mathcal{B}} = {}_{\mathcal{B}}[S]_{\mathcal{B}} {}_{\mathcal{B}}[T]_{\mathcal{B}}$$

$$T^n \mapsto {}_{\mathcal{B}}[T^n]_{\mathcal{B}} = ({}_{\mathcal{B}}[T]_{\mathcal{B}})^n$$

for all $S, T \in L(\mathcal{V}) := L(\mathcal{V}, \mathcal{V})$ and $n \in \mathbb{N}$.

- To study linear maps $T \in L(\mathcal{V})$ it is enough to study matrices.
- **Que:** Can we get basis $\mathcal{B}$ such that ${}_{\mathcal{B}}[T]_{\mathcal{B}}$ is “simple” in the sense that the matrix has more number of zero entries? For example, ${}_{\mathcal{B}}[T]_{\mathcal{B}}$ is a diagonal matrix.

Invariant subspaces

- **Definition:** A subspace $\mathcal{S} \subseteq \mathcal{V}$ is said to be **invariant** under $T \in L(\mathcal{V})$ if $T(x) \in \mathcal{S}$ for all $x \in \mathcal{S}$.

Eg: The subspaces $N(T)$ and $R(T)$ are invariant under T .

- Suppose $\mathcal{V} = \mathcal{S}_1 \oplus \mathcal{S}_2 \oplus \cdots \oplus \mathcal{S}_k$, where $\mathcal{S}_i$'s are subspaces of $\mathcal{V}$ with bases $\mathcal{B}_i$. Note that $\mathcal{B} := \cup_{i=1}^k \mathcal{B}_i$ is a basis for $\mathcal{V}$. Assume $\mathcal{S}_i$ is invariant under $T \in L(\mathcal{V})$ for every $1 \leq i \leq k$. Then

$${}_{\mathcal{B}}[T]_{\mathcal{B}} = \begin{bmatrix} A_1 & & & 0 \\ & A_2 & & \\ & & \ddots & \\ 0 & & & A_k \end{bmatrix}, \quad \text{where } A_i \in M_{d_i \times d_i}(\mathbb{F}), \quad d_i = \dim(\mathcal{S}_i).$$

- **Que:** How to get invariant subspaces $\mathcal{S}_i$? Can we choose them to be 1-dimensional so that ${}_{\mathcal{B}}[T]_{\mathcal{B}}$ becomes a diagonal matrix?
- **Observation:** Let $\mathcal{S} \subseteq \mathcal{V}$ be a 1-dimensional subspace invariant under T . Then there exists $0 \neq v \in \mathcal{S} \subseteq \mathcal{V}$ such that $\mathcal{S} = \text{span}\{v\} = \{\lambda v : \lambda \in \mathbb{F}\}$. Now $T(v) \in \mathcal{S}$ implies that $T(v) = \lambda v$ for some $\lambda \in \mathbb{F}$.

Eigenvalues and eigenvectors

- Suppose $\mathcal{V}$ is finite dimensional $\mathbb{F}$ -vector space and $T \in L(\mathcal{V})$.
- **Definition:** A scalar $\lambda \in \mathbb{F}$ is said to be an **eigenvalue** of $T \in L(\mathcal{V})$ if there exists $0 \neq v \in \mathcal{V}$ such that $T(v) = \lambda v$. Such a vector v is called an **eigenvector of T corresponding λ** . The subspace

$$\ker(T - \lambda I) = N(T - \lambda I) = \{v \in \mathcal{V} : T(v) = \lambda v\}$$

is called the **eigenspace of T corresponding to λ** .

- We let $\sigma(T)$ denotes the set of all eigenvalues of T including multiplicity.

- **Examples:**

1. $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T((x, y)^t) = (-y, x)^t$ has no eigenvalues.
2. $T : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ given by $T((x, y)^t) = (-y, x)^t$ has $\sigma(T) = \{i, -i\}$.

- **Exercises:**

1. Every $T \in L(\mathcal{V})$ has at most $\dim(\mathcal{V})$ eigenvalues.
2. If $\mathcal{V}$ is a **complex** vector space, then every $T \in L(\mathcal{V})$ has $\dim(\mathcal{V})$ eigenvalues (not necessarily distinct).
3. If $\mathcal{B}$ is a basis for $\mathcal{V}$, then $\sigma(T) = \sigma_{\mathcal{B}}[T]_{\mathcal{B}}$.
4. If $A \in M_n(\mathbb{F})$, then $\sigma(A) = \{\lambda \in \mathbb{F} : \det(A - \lambda I) = 0\}$.
5. Eigenvectors corresponding to distinct eigenvalues of T are linearly independent.
6. Show that eigenspaces of $T \in L(\mathcal{V})$ are invariant under T .

Diagonalizability

- **Definition:** $T \in L(\mathcal{V})$ is said to be **diagonalizable** if there exists a basis $\mathcal{B}$ for $\mathcal{V}$ such that ${}_{\mathcal{B}}[T]_{\mathcal{B}}$ is a diagonal matrix.
- **Recall:** If $\mathcal{B}, \tilde{\mathcal{B}}$ be two basis for $\mathcal{V}$, then ${}_{\tilde{\mathcal{B}}}[T]_{\tilde{\mathcal{B}}} = Q {}_{\mathcal{B}}[T]_{\mathcal{B}} Q^{-1}$, where $Q = {}_{\tilde{\mathcal{B}}}[I]_{\mathcal{B}}$. (Two matrices A, B are said to be **similar** if $B = Q A Q^{-1}$ for some invertible Q .)
- **Note:** Let $A \in M_n(\mathbb{F})$. Consider the linear map $T_A : \mathbb{F}^n \rightarrow \mathbb{F}^n$ given by $T_A(x) = Ax$ for all $x \in \mathbb{F}^n$. Then T_A is diagonalizable iff $Q A Q^{-1}$ is diagonal for some $Q \in M_n(\mathbb{F})$ invertible.
- **Definition:** A matrix $A \in M_{n \times n}(\mathbb{F})$ is said to be **diagonalizable** if there exists $P \in M_n(\mathbb{F})$ invertible and $D \in M_n(\mathbb{F})$ diagonal such that $A = P D P^{-1}$.
- $T \in L(\mathcal{V})$ is diagonalizable iff ${}_{\mathcal{B}}[T]_{\mathcal{B}}$ is diagonalizable for every basis $\mathcal{B}$ of $\mathcal{V}$.
- **Proposition:**
 1. If $T \in L(\mathcal{V})$ has $\dim(\mathcal{V})$ distinct eigenvalues, then T is diagonalizable.
 2. If $A \in M_n(\mathbb{F})$ has n distinct eigenvalues, then A is diagonalizable.
- **Que:** How to determine whether a given matrix/linear map is diagonalizable?

► **Recall:** Suppose $A \in M_n(\mathbb{F})$ and $\lambda \in \sigma(A)$.

- **algebraic multiplicity** (AM) of λ is its multiplicity as a root of the characteristic polynomial of A .
- **geometric multiplicity** (GM) of λ is the dimension of the eigenspace $\dim(N(T - \lambda I))$.
- $1 \leq GM(\lambda) \leq AM(\lambda)$.

► **Theorem:** Let $A \in M_n(\mathbb{F})$. Then the following conditions are equivalent:

1. A is diagonalizable.
2. $\mathbb{F}^n$ has a basis consisting of eigenvectors of A .
3. There exist one-dimensional subspaces $S_1, \dots, S_n$ of $\mathbb{F}^n$, each invariant under A , such that $\mathbb{F}^n = S_1 \oplus \dots \oplus S_n$.
4. $\mathbb{F}^n = N(A - \lambda_1 I) \oplus \dots \oplus N(A - \lambda_k I)$.
5. The sum of the geometric multiplicities of eigenvalues of A is equal to n .
6. The sum of algebraic multiplicities of the eigenvalues of A is equal to n , and for each eigenvalue, the geometric multiplicity equals algebraic multiplicity.

► **Exercises:** Check whether the following matrices are diagonalizable or not.

1. $\begin{bmatrix} 1/2 & 3/2 \\ 3/2 & 1/2 \end{bmatrix}$
2. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
3. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$
4. $\begin{bmatrix} 1 & 7 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & -1 \end{bmatrix}$
5. $\begin{bmatrix} 1 & 7 & 4 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$

Linear maps on inner product spaces

Suppose $\mathcal{V}, \mathcal{W}$ are $\mathbb{F}$ -vector spaces with basis $\mathcal{B}, \mathcal{E}$, respectively.

► Let $T \in L(\mathcal{V}, \mathcal{W})$ and $A = {}_{\mathcal{E}}[T]_{\mathcal{B}} \in M_{n \times m}(\mathbb{F})$.

► Consider ${}_{\mathcal{B}}[\cdot]_{\mathcal{E}} : L(\mathcal{W}, \mathcal{V}) \rightarrow M_{m \times n}(\mathbb{F})$.

$$?? \mapsto A^*$$

Need **innerproduct spaces (IPS)**.

► **Note:** Consider the linear map $T_A : \mathbb{F}^n \rightarrow \mathbb{F}^m$ by $T_A(x) = Ax$.

$\mathbb{F} = \mathbb{C}$:

$$\langle T_A(x), y \rangle = \langle Ax, y \rangle = \langle x, A^*y \rangle = \langle x, T_{A^*}(y) \rangle \quad \forall x \in \mathbb{C}^n, y \in \mathbb{C}^m.$$

$\mathbb{F} = \mathbb{R}$:

$$\langle T_A(x), y \rangle = \langle Ax, y \rangle = \langle x, A^t y \rangle = \langle x, T_{A^t}(y) \rangle \quad \forall x \in \mathbb{R}^n, y \in \mathbb{R}^m.$$

► **Theorem:** Let $\mathcal{V}, \mathcal{W}$ be **finite dimensional** inner product spaces and

$T \in L(\mathcal{V}, \mathcal{W})$. Then there exists a unique $S \in L(\mathcal{W}, \mathcal{V})$ such that

$$\langle T(v), w \rangle = \langle v, S(w) \rangle \quad \forall v \in \mathcal{V}, w \in \mathcal{W}.$$

This unique S is denoted by T^* and is called the **adjoint of T** .

► **Exercise:** Let $\mathcal{B}, \mathcal{E}$ be **orthonormal bases** for $\mathcal{V}, \mathcal{W}$, respectively. If $T \in L(\mathcal{V}, \mathcal{W})$ and $A = {}_{\mathcal{E}}[T]_{\mathcal{B}}$ then prove that $A^* = {}_{\mathcal{B}}[T^*]_{\mathcal{E}}$. What happens if the bases are not ONB?

Special class of linear transformation

► **Definition:** Let $\mathcal{V}, \mathcal{W}$ be finite dimensional inner product spaces and $T \in L(\mathcal{V}, \mathcal{W})$. Then T is said to be

1. **unitary** if $T^*T = I_{\mathcal{V}}$ and $TT^* = I_{\mathcal{W}}$.
2. **isometry** if $T^*T = I_{\mathcal{V}}$.

Further, if $\mathcal{V} = \mathcal{W}$, then T is said to be

3. **normal** if $T^*T = TT^*$.
4. **self-adjoint (Hermitian)** if $T^* = T$.
5. **projection (idempotent)** if $T^2 = T$.
6. **orthogonal projection** if $T^2 = T = T^*$.

► **Example:** $T : \mathbb{F}^2 \rightarrow \mathbb{F}^3$ given by $T((x_1, x_2)^t) = (x_1, x_2, 0)^t$ is an isometry but not unitary.

► **Examples:** The linear maps $T : \mathbb{F}^2 \rightarrow \mathbb{F}^2$ given by

1. $T((x_1, x_2)^t) = (x_1 + 2x_2, 2x_1 + 3x_2)^t$ is self-adjoint.
2. $T((x_1, x_2)^t) = (ix_2, -ix_1)^t$ is a unitary.
3. $T((x_1, x_2)^t) = (x_1 + x_2, 0)^t$ is a projection but **not** an orthogonal projection.
4. $T((x_1, x_2)^t) = (x_1, 0)^t$ is an orthogonal projection.

► **Exercise:** Suppose $\mathcal{V}, \mathcal{W}$ are complex finite dimensional IPS and let $T \in L(\mathcal{V}, \mathcal{W})$. Prove the following:

1. T is isometry $\iff \|T(v)\| = \|v\|$ for all $v \in \mathcal{V} \iff \langle T(v_1), T(v_2) \rangle = \langle v_1, v_2 \rangle$ for all $v_1, v_2 \in \mathcal{V}$.
2. T is unitary $\iff T$ is isometry and surjective (i.e., onto).
3. Suppose $\mathcal{V} = \mathcal{W}$. Then T is unitary $\iff T$ is isometry.

► **Remark:** Let $\mathcal{V}$ be a vector space $\mathcal{S}_1, \mathcal{S}_2 \subseteq \mathcal{V}$ be subspaces.

1. If $\mathcal{V} = \mathcal{S}_1 \oplus \mathcal{S}_2$, then given $v \in \mathcal{V}$ there exists unique $v_i \in \mathcal{S}_i$ such that $v = v_1 + v_2$. Define $P_i : \mathcal{V} \rightarrow \mathcal{V}$ by

$$P_1(v) := v_1, \quad (\text{note that } R(P_1) = \mathcal{S}_1, N(P_1) = \mathcal{S}_2.)$$

$$P_2(v) := v_2, \quad (\text{note that } R(P_2) = \mathcal{S}_2, N(P_2) = \mathcal{S}_1.)$$

The map $P_i \in L(\mathcal{V})$ is a projection, and is called the **projection of $\mathcal{V}$ onto $\mathcal{S}_i$** .

2. If $\mathcal{V}$ is an IPS and $\mathcal{V} = \mathcal{S}_1 \overset{\perp}{\oplus} \mathcal{S}_2$, then P_i is called the **orthonormal proj of $\mathcal{V}$ onto $\mathcal{S}_i$** .

► **Exercise:** Let $\mathcal{V}$ be a finite dimensional vector space and $P \in L(\mathcal{V})$.

1. If P is a projection, then $\mathcal{V} = R(P) \oplus N(P)$.
2. If $\mathcal{V}$ is an IPS and P is an orthogonal projection, $\mathcal{V} = R(P) \overset{\perp}{\oplus} N(P)$.
(Hint: If $v \in \mathcal{V}$, then $v = (I - P)v + Pv \in N(P) + R(P)$).

► **Exercise:** Let $\mathcal{V}$ be finite dimensional complex IPS and $T \in L(\mathcal{V})$.

1. $T \in L(\mathcal{V})$ is normal iff $\|Tv\| = \|T^*v\| \quad \forall v \in \mathcal{V}$.
2. Suppose T is normal. If $v \in \mathcal{V}$ is an eigenvector of T with eigenvalue λ , then v is also an eigenvector of T^* with eigenvalue $\bar{\lambda}$.
3. Suppose T is normal. Then eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

► **Cartesian decomposition:** Let $T \in L(\mathcal{V})$. Define

$$T_1 := \frac{T + T^*}{2} \quad \text{and} \quad T_2 = \frac{T - T^*}{2i}.$$

Note that T_i 's are self-adjoint such that $T = T_1 + iT_2$.

► **Theorem:** Let $\mathcal{V}, \mathcal{W}$ be finite dimensional complex inner product spaces with **orthonormal** bases $\mathcal{B}, \mathcal{E}$ respectively. Let $T \in L(\mathcal{V}, \mathcal{W})$ and $A := {}_{\mathcal{E}}[T]_{\mathcal{B}}$. Then

- (i) T is unitary iff $A^*A = I$ and $AA^* = I$.
- (ii) T is isometry iff $A^*A = I$.

Further, if $\mathcal{V} = \mathcal{W}$ and $\mathcal{B} = \mathcal{E}$, then

- (iii) T is normal iff $A^*A = AA^*$.
- (iv) T is self adjoint iff $A^* = A$.
- (v) T is projection (idempotent) iff $A^2 = A$.
- (vi) T orthogonal projection iff $A^2 = A = A^*$.

► **Proof:** Exercise.